

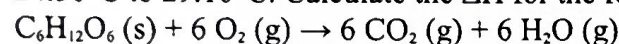
1. Multiple Choice Questions (18%)

1) The correct equation for which the enthalpy change defines the standard enthalpy of formation for $C_2H_5OH(l)$ is:

- A) $CH_2=CH_2(g) + H_2O(l) \rightarrow CH_3CH_2OH(l)$
 B) $CH_3CHO(l) + H_2(g) \rightarrow CH_3CH_2OH(l)$
 C) $4 C(gr) + 6 H_2(g) + O_2(g) \rightarrow 2 CH_3CH_2OH(l)$
 D) $2 C(gr) + 3 H_2(g) + 1/2 O_2(g) \rightarrow C_2H_5OH(l)$
 E) $2 C(gr) + 6 H(g) + O(g) \rightarrow C_2H_5OH(l)$

2) What is the difference between extensive and intensive properties? A) Extensive properties depend on the quantity of matter, while intensive properties do not. B) Extensive properties occur outside a system, while intensive properties occur inside. C) Extensive properties are state functions, while intensive properties are not. D) Extensive properties include color and smell, while intensive properties include enthalpy and mass. E) Extensive properties depend on the kinetic energy of a system, while intensive properties depend on the potential energy.

3) The heat capacity of a bomb calorimeter is found to be $5.68 \text{ kJ } ^\circ\text{C}^{-1}$. When a 1.652 g sample of glucose, $C_6H_{12}O_6$, is burned in the calorimeter, the temperature of the calorimeter increases from 24.56°C to 29.10°C . Calculate the ΔH for the following reaction:



- A) $-3.0 \times 10^3 \text{ kJ}$; B) -25.8 kJ ; C) 15.6 kJ ; D) -42.6 kJ ; E) 25.8 kJ /mol

4) Which of the following statements is incorrect? A) The more atoms in its molecules, the greater will be the entropy of a substance; B) If a process is spontaneous, the reverse process is nonspontaneous; C) The greater the degree of disorder in a system, the more negative is its entropy; D) The entropy of a pure, perfect crystal is zero at 0 K ; E) The entropy of the universe must always increase for a spontaneous process.

5) For which of the following processes would the entropy change be positive? A) $Hg(l) (45^\circ\text{C}) \rightarrow Hg(l) (25^\circ\text{C})$; B) $H_2O(l) \rightarrow H_2O(s)$; C) $He(g) (1 \text{ atm}) \rightarrow He(g) (10 \text{ atm})$; D) $I_2(s) \rightarrow I_2(aq)$; E) $Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$.

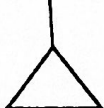
6) The reaction, $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$, has an equilibrium constant of 167.5 at 1000°C . What is the value of ΔG° for the reaction at 1000°C ? A) -42.6 ; B) -23.5 ; C) $-12.$; D) -54.2 kJ/mol .

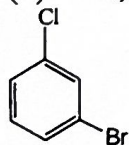
$$\Delta G^\circ = -RT \ln K_p$$

$$= -8.314$$

2. Name the following compounds, (30%)

(a) HCN (when you want to emphasis it is a gaseous compound); (b) $Cu_3(PO_4)_2$;

(c) SiH_4 ; (d) $K_2C_2O_4$; (e) $BaCl_2 \cdot 2H_2O$; (f) SO_4^{2-} ; (g) C_2H_2 ; (h) $HCOOH$; (i) ; (j)



3. Please complete the following table regarding the three thermodynamic system. (6%)

System	Open	Close	(isolated)
Heat	O	(X)	(X)
Matter	(O)	(X)	(X)

(Hint: O means the system allows exchange heat or matter with the surrounding; X means not allows)

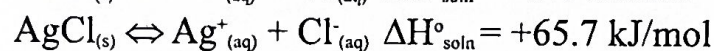
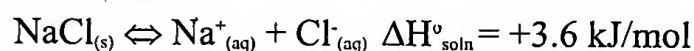
4. Zinc and HCl react according to the reaction, $Zn_{(s)} + 2HCl_{(aq)} \rightarrow Zn^{2+}_{(aq)} + 2Cl^{-}_{(aq)} + H_{2(g)}$. When 3.00 g of Zinc metal is added to a dilute HCl solution at 1.00 atm and 25°C ,

and the above reaction is allowed to go to completion at constant pressure, 13.98 kJ of heat must be removed to the final solution to its original temperature. What are the values of work and the change of internal energy? (Hint: MW = 65.409 g mol⁻¹ for Zn). (6%)

5. (a) Calculate the change in both enthalpy and entropy when a 3.0g block of ice melts at 0.0°C [$\Delta H_{\text{fus}}^{\circ}(\text{H}_2\text{O}) = 6.01 \text{ kJ/mol}$]. (b) For the same block of ice, calculate the entropy change for the system when the ice is warmed from 0.0°C to 25°C. The heat capacity of liquid water over this temperature range is 75.3 J/mol°C. (5%)

6. The oxidation of $\text{SO}_2(\text{g})$, $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$, is the key reaction in the manufacture of sulfuric acid. (a) Calculate the equilibrium constant K_p at 25°C. ($\Delta G_{298}^{\circ} = -141.6 \text{ kJ}$ for the reaction as written above) (b) Is this reaction (forward) spontaneous at 25°C? (c) A container is filled with 0.500 atm of SO_2 , 0.0100 atm of O_2 , and 0.100 atm of SO_3 and kept at 25°C. In which direction, if any, will the reaction process to reach equilibrium? (d) Calculate ΔG of the system in part (c). (e) What is the K_p at 102°C? [hint: $\Delta H^{\circ} = -198.4 \text{ kJ}$; $\Delta S^{\circ} = -187.9 \text{ JK}^{-1}$, for the reaction as written above] (f) Is this reaction spontaneous at 900°C? Why? (17%)

7. Consider the sample salts $\text{NaCl}_{(\text{s})}$ and $\text{AgCl}_{(\text{s})}$. The equilibria in which the two salts dissolve in water is examined:



(a) Calculate the values of ΔG° at 298K for each of the preceding reactions. (b) The two results in part (a) are very different. To understand the reasons, calculate the two entropy changes of the preceding reactions at 298 K. From the values, identify whether the enthalpy term or the entropy term is the primarily reason to the difference. (c) Use the ΔG° at 298K to calculate the K_{sp} values for the two salts. (d) How will the solution process of the salts change with the increasing temperature, assuming that ΔH and ΔS do not change with temperature? (Hint: $\Delta G^{\circ}_{\text{f}}(\text{Na}^+_{(\text{aq})}) = -261.9 \text{ kJ/mol}$; $\Delta G^{\circ}_{\text{f}}(\text{Cl}^-_{(\text{aq})}) = -131.2 \text{ kJ/mol}$; $\Delta G^{\circ}_{\text{f}}(\text{NaCl}_{(\text{s})}) = -384.0 \text{ kJ/mol}$; $\Delta G^{\circ}_{\text{f}}(\text{Ag}^+_{(\text{aq})}) = 77.11 \text{ kJ/mol}$; $\Delta G^{\circ}_{\text{f}}(\text{AgCl}_{(\text{s})}) = -109.7 \text{ kJ/mol}$; K_{sp} : the product of the concentrations of the ions) (18%)